Q1 (1) - Single Correct

We have, $10^{99} = 2^{99} \times 5^{99}$

Total number of divisor $= (99+1)(99+1) = (100)^2$

and $10^{88} = 2^{88} \times 5^{88}$

The multiple of 10^{88}, which is positive divisor of $10^{99} = 2^{11} \times 5^{11}$

$\therefore$ Total number of divisor of $2^{11} \times 5^{11} = (11+1)(11+1) = 12^2$

$\therefore$ Required probability $= \frac{(12)^2}{(100)^2} = \left(\frac{3}{25}\right)^2 = \frac{9}{625}$

where, $m = 9$ and $n = 625$ $m + n = 9 + 625 = 634$

Hence, option (1) is correct.

Q2 (4) - Single Correct

Total number of determinant is formed of order 2 taken 0 and 1 element only $= 2^4 = 16$

There are three possible cases for determinant have positive

value $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\therefore$ Required probability $= 1 - \frac{3}{16} = \frac{13}{16}$

Hence, option (4) is correct.

Q3 (3) - Single Correct

Let the x and y be the black and red colour balls, respectively.

Now, $\quad x + y = 10$

and $\frac{{}^XC_1 \times {}^yC_1}{(x+y)'C_2} = \frac{8}{15} = \frac{xy}{{}^{10}C_2} = \frac{8}{15} \quad [\because x + y = 10]$

$= xy = \frac{8 \times 45}{15}$

$\Rightarrow \quad xy = 24$ and $x + y = 10$

On solving these equations, we get

$x = 6, y = 4$ or $y = 6$ and $x = 4$

Now, here black ball is more than red balls.

$\therefore x = 6$ and $y = 4$

$\therefore$ Required probability $= \frac{{}^6C_1 \times {}^4C_1}{{}^{10}C_2} + \frac{{}^6C_2}{{}^{10}C_2}$

$= \frac{6 \times 4}{45} + \frac{6 \times 5}{10 \times 9}$

$= \frac{24+15}{45} = \frac{39}{45}$

Hence, option (3) is correct.

Q4 (2) - Single Correct

Let $P(S)=P(1 \text{ or } 2)=\frac{1}{3}$

$P(F)=P(3 \text{ or } 4 \text{ or } 5 \text{ or } 6)=\frac{2}{3}$

$P(A \text{ wins })=P[SS$ or SFSS or SFSFSS or ...) or (FSS or FSFSS or ...)]

$=\frac{\frac{1}{9}}{1-\frac{2}{9}}+\frac{\frac{2}{27}}{1-\frac{2}{9}}$

$=\frac{1}{9} \times \frac{9}{7}+\frac{2}{27} \times \frac{9}{7}$

$=\frac{5}{21}$

Q5 (1) - Single Correct

We have, $P(A)=P(7)=\frac{6}{36}$

and $P(B)=P(4)=\frac{3}{36}$

Since, equal throws are disregarded.

Hence, in each throw A is twice as likely to win as B.

Let $P(B)=\rho$

$P(A)=2p$

$\therefore 3p=1$

$\Rightarrow p=\frac{1}{3}$

Q6 (2) - Single Correct

We have, $\{1,2,3,4,5\}$ Total number of selection a, b, c from $\{1,2,3,4,5,\}$, where a, b, c are different or may not be different is 5^3 i.e. 125

Now, $ab+c$ is even. If ab and c both are odd or even.

a	b	c	Total number of ways
Odd	Odd	Odd	${}^3C_1 \times {}^3C_1 \times {}^3C_1=27$
Odd	Even	Even	${}^3C_1 \times {}^2C_1 \times {}^2C_1=12$
Even	Odd	Even	${}^2C_1 \times {}^3C_1 \times {}^2C_1=12$
Even	Even	Even	${}^2C_1 \times {}^2C_1 \times {}^2C_1=8$

$$\therefore \text{ Required probability } =\frac{27+12+12+8}{125}$$
$$=\frac{59}{125}$$

Hence, option (2) is correct.

Q7 (3) - Single Correct

Let E_1 be the event Mr. A transfer one ball to Mr. B and E_2 be the event the same colour ball transfer to Mr. A.

Now, $P\left(E_1\right)=\frac{1}{5}$

$P\left(E_2\right)=\frac{2}{6}$

Now, this process is repeated five times.

$$\begin{aligned} \text { Required probability } & =5 \times P\left(E_1\right) \times P\left(E_2\right) \\ & =5 \times \frac{1}{5} \times \frac{2}{6}=\frac{1}{3} \end{aligned}$$

Hence, option (3) is correct.

Q8 (3) - Single Correct

Here, there are n ropes lying on the floor.

Let two boys A and B anyone of n ropes hold.

$\therefore$ Total number of outcomes $=n+n=2n$

Required probability $\frac{n}{{ }^n C_2}=\frac{1}{101}$

$\Rightarrow \quad \frac{n \times 2}{2 n(2 n-1)}=\frac{1}{101}$

$\therefore \quad n=51$

Hence, option (3) is correct.

Q9 (3) - Single Correct

We have, word 'STATISTICS' have S occur 3 times, Toccur 3 times, 1 occur 2 times and A and C are different, and the word 'ASSISTANT' have S occurs 3 times, T occurs 2 times A occurs 2 times and I, and N different. Proabability of S, T, I, A, C from word 'STATISTICS' are $\frac{3}{10}, \frac{3}{10}, \frac{2}{10}, \frac{1}{10}, \frac{1}{10}$ respectively and probability of S, T, A, I and N from word 'ASSISTANT' are $\frac{3}{9}, \frac{2}{9}, \frac{2}{9}, \frac{1}{9}, \frac{1}{9}$ respectively. In the given 'STATISTICS' and 'ASSISTANT' have same letters are S, T. and A.

$$\begin{aligned} \therefore \text { Required probability } & =\frac{3}{10} \times \frac{3}{9}+\frac{3}{10} \times \frac{2}{9}+\frac{2}{10} \times \frac{2}{9} \\ & =\frac{9+6+4}{90}=\frac{19}{90} \end{aligned}$$

Hence, option (3) is correct.

Q10 (2) - Single Correct

If A fair coin is tossed $(m+n)$ times.

The probability of atleast m times consecutive head occurs is $\frac{n+2}{2^{m+1}}$.

Here, $n=4$ and $m=3$

$\therefore$ Required probability $=\frac{4+2}{2^{3+1}}=\frac{6}{2^4}=\frac{6}{16}=\frac{3}{8}$

Hence, option (2) is correct.

Q11 (3) - Single Correct

Put $\tan 2x=t, -\infty<t<\infty$

$\left[t^2\right]=t+a, a \in N$

So, t should be an integer.

$\Rightarrow\left(t^2\right)-t-a=0$

$\therefore t=\frac{1 \pm \sqrt{1+4a}}{2}$

Here, $1+4a$ is odd number and which is perfect square.

$1+4a=9,25, \ldots,(63)^2$

$\therefore$ Number of values that a can take $=31$

$\Rightarrow p=\frac{31}{1000}$

Q12 (3) - Single Correct

Total number of ways of selecting three numbers from $\{1,2,3, \ldots, n\}$ with replacement $=n \times n \times n=n^3$

Now, let us find the favourable cases. Let $x+y+z=2n$, where $1 \leq x \leq n, 1 \leq y \leq n$ and $1 \leq z \leq n$. Clearly, favourable number of cases is the number of solution of above equation

$=$ Coefficient of x^{2n} in $\left(x+x^2+\cdots+x^n\right)^3$

$=$ Coefficient of x^{2n} in $x^3\left(1+x+x^2+\cdots+x^{n-1}\right)^3$

$=$ Coefficient of x^{2n-3} in $\left(1+x+x^2+\cdots+x^{n-1}\right)^3$

$=$ Coefficient of x^{2n-3} in $\left(\frac{1-x^n}{1-x}\right)^3$

$=$ Coefficient of x^{2n-3} in $\left(1-x^n\right)^3(1-x)^{-3}$

$=$ Coefficient of x^{2n-3} in $\left(1-x^{3n}-3x^n+3x^{2n}\right)(1-x)^{-3}$

$=$ Coefficient of x^{2n-3} in $(1-x)^{-3}+(-3)$

$=$ Coefficient of x^{n-3} in $(1-x)^{-3}$

$={ }^{2n-1}C_2-3^{n-1}C_2=\frac{(n-1)(n-4)}{2}$

$\therefore$ Required probability $=\frac{(n-1)(n+4)}{2n^3}$

Q13 (3, 4) - Multiple Correct

Here, $E_1=\{5,10,15,20,25,\ldots\}$ and $E_2=\{5,15,25,35\ldots\}$

(1) E_1 is not twice as likely to ocaur E_2.

Hence, statement is incorrect.

(2) $E_1\cap E_2=\{5,15,25,\ldots\}$

Hence, this statement is incorrect.

(3) Here, $P\left(E_1\right)=1-\left(\frac{8}{10}\right)^2=\frac{36}{100}$

and $\quad P\left(E_2\right)=1-\left(\frac{9}{10}\right)^2=\frac{9}{100}$

$\therefore \quad P\left(E_1\cap E_2\right)=\frac{9}{100}$

$P\left(\frac{E_2}{E_1}\right)=\frac{\frac{9}{100}}{\frac{36}{100}}=\frac{1}{4}$

Hence, this statement is correct.

(4) $P\left(\frac{E_1}{E_2}\right)=\frac{\frac{9}{100}}{\frac{9}{100}}=1$

Hence, this statement is correct.

Q14 (1, 2, 3, 4) - Multiple Correct

(1) $P\left(\frac{A}{B^c}\right)=\frac{P(A\cap B^c)}{P(B^c)}=\frac{P(A)-P(A\cap B)}{1-P(B)}$

(2) we have, $P(A\cup B)\leq 1$

$\Rightarrow P(A)+P(B)-P(A\cap B)\leq 1$

$\Rightarrow P(A\cap B)\geq P(A)+P(B)-1$

(3) we have, $\quad P\left(\frac{A}{B^c}\right)<P(A)$

$\Rightarrow \quad \frac{P(A\cap B^c)}{P(B^c)}<P(A)$

$\Rightarrow \quad \frac{P(A)-P(A\cap B)}{1-P(B)}<P(A)$

$\Rightarrow \quad P(A)-P(A\cap B)<P(A)-P(A)P(B)$

$\Rightarrow \quad P(A)P(B)<P(A\cap B)$

$\Rightarrow \quad P(A)\times P(B)<P(B)\times P\left(\frac{A}{B}\right)$

$\Rightarrow \quad P(A)<P\left(\frac{A}{B}\right)$

(4) $P\left(\frac{A}{B^c}\right)+P\left(\frac{A^c}{B^c}\right)=\frac{P(A\cap B^c)}{P(B^c)}+\frac{P(A^c\cap B^c)}{P(B^c)}$

$=\frac{P(A)-P(A\cap B)+1-P(A\cup B)}{P(B^c)}$

$=\frac{P(A)+1-[P(A\cup B)+P(A\cap B)]}{P(B^c)}$

$=\frac{P(A)+1-P(A)-P(B)}{P(B^C)}$

$=\frac{1-P(B)}{P(B^c)}=\frac{P(B^c)}{P(B^c)}=1$

Q15 (2, 3, 4) - Multiple Correct

(1) $P(TTT \text{ or } HHH)=\frac{1}{8}+\frac{1}{8}=\frac{1}{4}$

Hecne, this statement is incorrect.

(2) $\frac{P(A\cap B)}{P(B)}=\frac{P(A\cap\bar{B})}{P(B)}=\frac{P(A)-P(A\cap B)}{1-P(B)}$

$\Rightarrow \quad P(A\cap B)[1-P(B)]=P(B)[P(A)-P(A\cap B)]$

$\Rightarrow P(A\cap B)-P(A\cap B)P(B)=P(A)P(B)-P(B)P(A\cap B)$

$\Rightarrow \quad P(A\cap B)=P(A)\cdot P(B)$

Since, A and B are independent events. Hence, this statement is correct.

(3) Let E_1 be the white ball is drawn in first drawn, E_2 be the event that black is drawn in second drawn and E be the event that white ball is drawn in second drawn.

$$\begin{aligned}\therefore P(E)&=P(E_1)\times P\left(\frac{E}{E_1}\right)+P(E_2)\times\left(\frac{E}{E_2}\right)\\&=\left(\frac{w}{w+b}\right)\left(\frac{d+w}{w+b+d}\right)+\frac{b}{(w+b)}\left(\frac{w}{w+b+d}\right)\\&=\frac{w(d+w+b)}{(w+b)(w+b+d)}=\frac{w}{w+b}\end{aligned}$$

which is independent of d. Hence, this statement is correct

(4) To prove that A,B,C are pairwise independent only. Now,

$$\begin{aligned}P(A\cap B)&=P[(A\cap B\cap\bar{C})\cup P(A\cap B\cap C)]\\&=P(A\cap B\cap\bar{C})+P(A\cap B\cap C)\\&=P(A)P(B)P(\bar{C})+P(A)P(B)P(C)\\&=P(A)\times P(B)[P(\bar{C})+P(C)]\\&=P(A)\times P(B)\end{aligned}$$

Hence, this statement is correct.

Q16 (3) - Multiple Correct

If S_1 and S_2 are in the same pair, then exactly one wins. The number of ways 16 players are divided into 8 pairs is

$n(S)=\frac{16!}{(2!)^8\times 8!}$

The number of ways in which S_1 and S_2 are in same pairs.

$n(E)=\frac{14!}{(2!)^7\times 7!}$

$\therefore$ The probability of S_1 and S_2 in the same pairs $=\frac{\frac{14!}{(2!)^7\times 7!}}{\frac{16!}{(2!)^8\times 8}}=\frac{1}{15}$

The probability of anyone winning in the pair of S_1S_2 is P (certain event) $=1$ Hence, the probability that the pair of S_1,S_2 being in two pairs separately and anyone of S_1,S_2 wins is given by the probability S_1,S_2 being in two pairs separately and S_1 wins, S_2 loses $+$ the probability of S_1,S_2 being in two pairs separately and S_1 loses, S_2 wins, it is given by

$\left(1-\frac{1}{15}\right) \times \frac{1}{2} \times \frac{1}{2}+\left(1-\frac{1}{15}\right) \times \frac{1}{2} \times \frac{1}{2}=\frac{7}{15}$

Required probability $=\frac{1}{15}+\frac{7}{15}=\frac{8}{15}$

Q17 (4) - Multiple Correct

3^a ends in $\rightarrow$ / 7^b ends in $\downarrow$	1	3	7	9
1			8	
3				
7	8			
9				8

Out of 16 cases, 3 are favourable.

$\Rightarrow \quad P=\frac{3}{16}$

Q18 (3) - Multiple Correct

Consider the events E_1, E_2, E_i and G.

$E_1=A$ receives the signal correct

$E_2=B$ receives the signal correct

$E_i=$ Signal received by B is green

$G=$ Original signal is green

$P\left(E_1\right)=\frac{3}{4}, P\left(E_2\right)=\frac{3}{4}$

$P\left(\bar{E}_1\right)=\frac{1}{4}, \quad P\left(\bar{E}_2\right)=\frac{1}{4}$

$P(G)=\frac{4}{5}, \quad P(\bar{G})=\frac{1}{5}$

$P(E)=P\left(G E_1 E_2\right)+P\left(G \bar{E}_1 \bar{E}_2\right)+P\left(\bar{G} E_1 \bar{E}_2\right)+P\left(\bar{G} \bar{E}_1 E_2\right)$

$$P(E)=\frac{4}{5} \times \frac{3}{4} \times \frac{3}{4}+\frac{4}{5} \times \frac{1}{4} \times \frac{1}{4}+\frac{1}{5} \times \frac{3}{4} \times \frac{1}{4}+\frac{1}{5} \times \frac{1}{4} \times \frac{3}{4}$$
$$=\frac{36+4+3+3}{80}=\frac{46}{80}$$

Required probability,

$P(G / E)=\frac{P\left(G E_1 E_2\right)+P\left(G \bar{E}_1 \bar{E}_2\right)}{P(E)}=\frac{\frac{36}{80}+\frac{4}{80}}{\frac{46}{80}}=\frac{20}{23}$

Q19 (2, 3, 4) - Multiple Correct

Let number of blue marbles in b and number of green marbles is g.

Hence $\frac{b g}{{ }^{b+g} C_2}=\frac{1}{2}$

$\Rightarrow \quad (b+g)(g+b-1)=4bg$

$\Rightarrow \quad (b+g)^2-(b+g)=4bg$

$\Rightarrow \quad b^2+g^2+2bg-b-g=4bg$

$\Rightarrow \quad g^2-2bg-g+b^2-b=0$

$\Rightarrow \quad g^2-(2b+1)g+\left(b^2-b\right)=0$

$D=(2b+1)^2-4\left(b^2-b\right)$

$\quad =(8b+1)$ must be a perfect square

Hence, possible values of b are $3, 6, 10$.

Q20 (1) - Multiple Correct

R : They obtained the same result

$B_1 : A \cap \bar{B}; P\left(B_1\right)=\frac{1}{8} \cdot \frac{11}{12}$

$B_2 : \bar{A} \cap B; P\left(B_2\right)=\frac{7}{8} \cdot \frac{1}{12}$

$B_3 : A \cap B; P\left(B_3\right)=\frac{1}{8} \cdot \frac{1}{12}$

$B_4 : \bar{A} \cap \bar{B}; P\left(B_4\right)=\frac{7}{8} \cdot \frac{11}{12}$

Now, $P\left(R/B_1\right)=0, P\left(R/B_2\right)=0, P\left(R/B_3\right)=1$

$$P\left(R/B_4\right)=\frac{7}{8} \cdot \frac{11}{12} \cdot \frac{1}{1001}$$

$$P\left(B_3/4\right)=\frac{\frac{1}{8} \cdot \frac{1}{12}}{\frac{1}{8} \cdot \frac{1}{12}+\frac{7}{8} \cdot \frac{11}{12} \cdot \frac{1}{1001}}=\frac{13}{14}$$

Q21 (3) - Multiple Correct

Let E_1, E_2, E_3 and A be the events defined as follow:

E_1 : The bolt is manufactured by machine A.

E_2 : The bolt is manufactured by machine B.

E_3 : The bolt is manufactured by machine C.

A: The bolt is defective.

Now, $P\left(E_1\right)=\frac{35}{100}=\frac{7}{20}, \quad P\left(E_2\right)=\frac{25}{100}=\frac{1}{4}$

$P\left(E_3\right)=\frac{40}{100}=\frac{4}{10}=\frac{2}{5}$

$P\left(\frac{A}{E_1}\right)$: Probability that the bolt drawn in defective given that it is manufactured by machine $A=\frac{2}{100}=\frac{1}{50}$

Similarly, $P\left(\frac{A}{E_2}\right)=\frac{1}{100}$

$P\left(\frac{A}{E_3}\right)=\frac{3}{100}$

By Baye's theorem, we have

$$P\left(\frac{E_1}{A}\right)=\frac{P\left(E_3\right)P\left(\frac{A}{E_3}\right)}{P\left(E_1\right)P\left(\frac{A}{E_1}\right)+P\left(E_2\right)P\left(\frac{A}{E_2}\right)+P_{\{}E_3\big)\rho\left(\frac{A}{E_3}\right)}$$

$$=\frac{\frac{2}{5}\times\frac{3}{100}}{\frac{7}{20}\times\frac{1}{50}+\frac{1}{4}\times\frac{1}{100}+\frac{2}{5}\times\frac{3}{100}}$$

$$=\frac{\frac{6}{500}}{\frac{7}{1000}+\frac{1}{400}+\frac{6}{500}}=\frac{\frac{6}{500}}{\frac{14+5+24}{2000}}=\frac{\frac{6}{500}}{\frac{43}{2000}}$$

$$=\frac{6}{500}\times\frac{2000}{43}=\frac{24}{43}$$

Q22 (1, 2) - Multiple Correct

Given, $P(X/Y)=\frac{1}{2}$ and $P(Y/X)=\frac{1}{3}$

and $\quad P(X\cap Y)=\frac{1}{6}$

Now $\quad P(X)=\frac{P(X\cap Y)}{p(Y/X)}$

$\Rightarrow \quad P(X)=\frac{1/6}{1/3}=\frac{1}{6}\times\frac{3}{1}=\frac{1}{2}$

and $\quad P(Y)=\frac{P(X\cap Y)}{P(X/Y)}=\frac{1/6}{1/2}=\frac{1}{6}\times\frac{2}{1}=\frac{1}{3}$

$P(X/Y)=P(X)=\frac{1}{2}$

and $\quad P(Y/X)=P(Y)=\frac{1}{3}$

$\therefore X$ and Y are independent events.

Now, $P(X\cup Y)=P(X)+P(Y)-P(X\cap Y)$

$=P(X)+P(Y)-P(X)\cdot P(Y)$

$=\frac{1}{2}+\frac{1}{3}-\frac{1}{2}\times\frac{1}{3}$

$=\frac{5}{6}-\frac{1}{6}=\frac{4}{6}=\frac{2}{3}$

and $P(\bar{X}\cap Y)=P(Y)-P(X\cap Y)$

$=\frac{1}{3}-\frac{1}{6}=\frac{1}{6}$

Q23 (1, 4) - Multiple Correct

Let $P(E)=e$ and $P(F)=f$

$\therefore \quad P(\bar{E})=(1-e)$ and $P(\bar{F})=(1-f)$

Now, $\quad P(E\cup F)-P(E\cap F)=\frac{11}{25}$

$\Rightarrow \quad e+f-2ef=\frac{11}{25} \quad \ldots$(i)

and $P(\bar{E}\cap\bar{F})=\frac{2}{25}$

$\Rightarrow (1-e)(1-f)=\frac{2}{25}$

$\Rightarrow 1-e-f+ef=\frac{2}{25} \quad \ldots$(ii)

[$\because E$ and F are independent, $\bar{E}$ and $\bar{F}$ are also independent, i.e.

$P(\bar{E} \cap \bar{F}) = P(\bar{E}) \cdot P(\bar{F})]$

From E as. (i) and (ii), we have

$ef = \frac{12}{25}$ ...(iii)

and $e + f = \frac{7}{5}$ ...(iv)

On solving Eqs. (iii) and (iv), we get

$e\left(\frac{7}{5} - e\right) = \frac{12}{25}$

$\Rightarrow \quad e^2 - \frac{7}{5}e + \frac{12}{25} = 0$

$\Rightarrow \quad \left(e - \frac{3}{5}\right)\left(e - \frac{4}{5}\right) = 0$

$\therefore \quad e = \frac{3}{5}$ or $\frac{4}{5}$

$\Rightarrow \quad f = \frac{4}{5}$ or $\frac{3}{5}$

Q24 (2) - Paragraph 1

We have, b red balls, $2b$ white balls and $3b$ blue balls.

$\therefore$ Total number of balls $= b + 2b + 3b = 6b$

$A =$ The event have no two of the selected balls are same colour

i.e. all are different

$\therefore P(A) = \frac{{}^{b}C_1 \times {}^{2b}C_1 \times {}^{3b}C_1}{{}^{6b}C_3}$

$= \frac{b \times 2b \times 3b \times 1 \times 2 \times 3}{6b(6b-1)(6b-2)}$

$= \frac{6b^2}{36b^2 - 18b + 2}$

Now, $\quad P(A) = 0.3$

$\therefore \quad \frac{6b^2}{36b^2 - 18b + 2} = 0.3 = \frac{3}{10}$

$\Rightarrow \quad 20b^2 = 36b^2 - 18b + 2$

$\Rightarrow \quad 16b^2 - 18b + 2 = 0$

$\Rightarrow \quad 8b^2 - 9b + 1 = 0$

$\Rightarrow \quad 8b^2 - 8b - b + 1 = 0$

$\Rightarrow \quad (8b - 1)(b - 1) = 0$

$b = 1$

So, there is exactly one value of b for which $P(A) = 0.3$ this is less than 10

Hence, option (2) is correct.

Q25 (4) - Paragraph 1

Required probability

$= \frac{3b^{3b}C_3}{^{6b}C_3} + \frac{^{3b}C_1{}^{3b}C_2}{^{6b}C_3}$
$= \frac{3b(3b-1)(3b-2)}{6b(6b-1)(6b-2)} + \frac{3b(3b)(3b-1)\times 3}{6b(6b-1)(6b-2)}$
$= \frac{3b(3b-1)(3b-2+9b)}{6b(6b-1)(6b-2)}$
$= \frac{3b(3b-1)(12b-2)}{6b(6b-1)(6b-2)} = \frac{1}{2}$

Hence, option (4) is correct.

Q26 (3) - Paragraph 2

We have, A_1 have one balls $\{1\}$

A_2 have two balls $\{1,2\}$

A_3 have three balls $\{1,2,3\}$

A_4 have four balls $\{1,2,3,4\}$

$P(A_1) = \frac{1}{10}, \quad P(A_2) = \frac{2}{10}, \quad P(A_3) = \frac{3}{10}, \quad P(A_4) = \frac{4}{10}$

$P\left(\frac{E_1}{A_1}\right) = 1, \quad P\left(\frac{E_2}{A_2}\right) = \frac{1}{2}, \quad P\left(\frac{E_3}{A_3}\right) = \frac{1}{3}, \quad P\left(\frac{E_4}{A_4}\right) = \frac{1}{4}$

$\therefore \sum_{i=1}^{4} P(E_i \cap A_i) = P(E_1 \cap A_1) + P(E_2 \cap A_2) + P(E_3 \cap A_3) + P(E_4 \cap A_4)$

$= P(A_1) \times P\left(\frac{E_1}{A_1}\right) + P(A_2) \times P\left(\frac{E_2}{A_2}\right) + P(A_3) \times P\left(\frac{E_3}{A_3}\right) + P(A_4) \times P\left(\frac{E_4}{A_4}\right)$

$= \frac{1}{10} \times 1 + \frac{2}{10} \times \frac{1}{2} + \frac{3}{10} \times \frac{1}{3} + \frac{4}{10} \times \frac{1}{4} = \frac{2}{5}$

Hence, option (3) is correct.

Q27 (2) - Paragraph 2

$P\left(\frac{A_3}{E_2}\right) = \frac{P(A_3)\times P\left(\frac{E_2}{A_3}\right)}{\left[P(A_1)\times P\left(\frac{E_2}{A_1}\right)+P(A_2)\times P\left(\frac{E_2}{A_2}\right)+P(A_3)\times P\left(\frac{E_2}{A_3}\right)+P(A_4)\times P\left(\frac{E_2}{A_4}\right)\right]}$

$= \frac{\frac{3}{10}\times\frac{1}{3}}{\frac{1}{10}\times 0+\frac{2}{10}\times\frac{1}{2}+\frac{3}{10}\times\frac{1}{3}+\frac{4}{10}\times\frac{1}{4}} = \frac{1}{3}$

Hence, option (2) is correct.

Q28 (1) - Paragraph 3

Here, $P_n = P_{n-1} + P_{n-2}$ and $P_1 = 1, P_2 = 1$

So, $P_3 = 2, P_4 = 3$

Q29 (2) - Paragraph 3

Using, $P_n = P_{n-1} + P_{n-2}$

$\therefore \quad P_8 = 21$

Q30 (2) - Paragraph 4

Total number of balls $=$ (4 white $+9$ black $)=13$ balls

Total number of selection of 13 balls with replacement $=13^4$

$\therefore$ Required probability

$P(BBBB)+P(BBBW)+P(BBWB)+P(BWBB)+P(WBBB)+P(WBBW)+P(WBWB)+P(BWBW)$
$=\left(\frac{9}{13}\right)^4+\frac{9^3\times 4}{13^4}+\frac{9^3\times 4}{13^4}+\frac{9^3\times 4}{13^4}+\frac{9^3\times 4}{13^4}\frac{9^2\times 4^2}{13^4}+\frac{9^2\times 4^2}{13^4}+\frac{9^2\times 4^2}{13^4}$
$=\frac{9^2}{13^4}(81+36+36+36+36+16+16+16)$
$=\frac{9^2}{13^4}(273)=\frac{81\times 273}{(169)^2}$
Hence, option (2) is correct.

Q31 (2) - Paragraph 4

Recursion relation of p(r) must be $P_{(r)}=P_{(r-2)}\cdot\left(\frac{9}{13}\right)^2+P_{(r+1)}\left(\frac{9}{13}\right)$

Q32 (2) - Paragraph 5

Total number of selection of set of students $\{a_1, a_2, a_3, \ldots, a_n\}$ plans to visit two institutes next month $=2^n\times 2^n=4^n$

Suppose, r students visit 1 st institute, where r varies from 0 to n.

$\therefore$ Total number of selection $={}^n$C, ways. So, $(n-r)$ student visit 2nd institute, because 1 st and 2nd institute have no common student.

$\therefore$ Total number of selection for $n-r=2^{n-r}$ ways.

The total number of student visit 1 st and 2 nd institute is chosen in ${}^nC_r 2^{n-r}$ ways.

But r can varies from 0 to n.

$\therefore \sum_{r=0}^n {}^nC_r 2^{n-r}=(1+2)^n=3^n$

Now, required probability $=\frac{3^n}{4^n}=\left(\frac{3}{4}\right)^n$

Hence, option (2) is correct.

Q33 (3) - Paragraph 5

Suppose, r and $n-r$ students visit 1 st and 2 nd institute, respectively. Any number of student chosen from r student, so total number of student selecting from either in r and $n-r={}^nC_r 2^r$

But r can varie from 0 to n.

$\therefore$ Total number of ways for selecting $=\sum_{r=0}^n {}^nC_r 2^{\prime}$

$=(1+2)^n=3^n$

$\therefore$ Hence, required probability $= \frac{3^n}{4^n} = \left(\frac{3}{4}\right)^n$

Hence, option (3) is correct.

Q34 (3) - Paragraph 6

$P(E) = \frac{{}^9C_3\left[{}^3C_1 \cdot {}^2C_2 \cdot \frac{5!}{3!} + {}^3C_2 \cdot {}^1C_1 \cdot \frac{5!}{2!2!}\right]}{9^5}$

$= \frac{{}^9C_3(60+90)}{9^5}$

$= \frac{{}^9C_3 \times 150}{9^5}$

Q35 (2) - Paragraph 6

Favourable $= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Here, $a_{11} = a_{22} = a_{33} = 0$

a_{12}, a_{13}, a_{23} can be filled in 9 ways each a_{21}, a_{31}, a_{32} has one way.

$\therefore n(E) = 9^3$

and $\quad n(S) = 9^9$

$\therefore \quad P(E) = \frac{n(E)}{n(S)}$

$= \frac{9^3}{9^9} = \frac{1}{9^6}$

Q36 (3) - Integer Type

Let, A be the even of choosing 5 -digits number and E be the event the number is a palindrome (abcba).

$\therefore n(A) = 9\left(10^4\right) = 90000$

and $n(A \cap B) = 9 \times 10^2 = 900$

$\therefore P(B/A) = \frac{n(B \cap A)}{n(A)} = \frac{1}{100} = p$

$300p = 3$

Q37 (8) - Integer Type

Let n pairs of socks in drawer.

$\therefore$ Total number of socks in drawer $= 2n$

Total number of out comes when two socks are drawn $= {}^{2n}C_2$

Favourable case $= {}^nC_1$

$\therefore$ Required probability $= \frac{{}^nC_1}{{}^{2n}C_2} = \frac{1}{(2n-1)}$

Now, $\frac{1}{2n-1}=\frac{1}{15}$

$\Rightarrow n=8$

Hence, 8 pairs of socks in drawer.

Q38 (6) - Integer Type

Total number of ways in which papers of 4 students, can be checked by seven teachers $=7^4$

Now, choosing two teachers out of $7={}^7C_2=21$

The number of ways in which 4 papers can be checked by exactly two teachers $=2^4-2=14$

$\therefore$ Favourable ways $=21\times 14$

Required probability $=\frac{21\times 14}{7^4}=\frac{6}{49}$

Now, $49A=49\times\frac{6}{49}=6$

Q39 (4) - Integer Type

The total number of mapping is n^n.

The number of one-one mapping is $n!$

Required probability $=\frac{n!}{n^n}=\frac{3}{32}=\frac{6}{64}=\frac{24}{256}$

$\Rightarrow \quad \frac{n!}{n^n}=\frac{4!}{4^4}$

On comparing, we get $n=4$

Q40 (5) - Integer Type

We have, n identical red balls, $2n$ identical black balls and $3n$

identical white balls. Total number of selection of n balls from $n, 2n$ and $3n$ balls,

$x_1+x_2+x_3=n$

$=$ Coefficient of $x^n\left(1+x+x^2+\cdots+x^n\right)$

$\left(1+x+x^2+\cdots+x^{2n}\right)\left(1+x+\cdots+x^{3n}\right)$

$=$ Coefficient of x^n in $\left(1-x^{n+1}\right)\left(1-x^{2n+1}\right)\left(1-x^{3n+1}\right)(1-x)^{-3}$

$=$ Coefficient of x^n in $(1-x)^{-3}={}^{n+3-1}C_{3-1}={}^{n+2}C_2$

Favourable case $=1+1+1=3$

$\therefore$ Required probability $=\frac{3}{{}^{n+2}C_2}$

$\therefore \quad \frac{6}{(n+2)(n+1)}\geq\frac{1}{6}$

$\Rightarrow \quad n^2+3n-34\geq 0$

$n=\frac{-3\pm\sqrt{9+136}}{2}=\frac{-3\pm\sqrt{145}}{2}$

$= \frac{-3 \pm 12.04}{2} > 4.56$

$\therefore \quad n \geq 5$

Q41 (6) - Integer Type

We have,

$p =$ Probability of getting head on flipping a coin

$(1-\rho) =$ Probability of getting tail on flipping a coin

Let $n =$ Number of times coin flipped $= 8$

$x =$ Number of times getting head Clearly $X \sim B(8, p)$

$\therefore \quad P(X = x) = {}^8C_{\times} p^x (1-p)^{8-x}, x = 0, 1, 2, \cdots, 8$

Now, we have

P (getting 3 head and five tails) $= \frac{1}{25} P($ getting 5 heads and 3 tails$)$

$\Rightarrow \quad P(X = 3) = \frac{1}{25} P(X = 5)$

$\Rightarrow \quad {}^8C_3 P^3 (1-p)^5 = \frac{1}{25} {}^8C_5 p^5 (1-\rho)^8 - 5$

$\Rightarrow \quad {}^8C_3 \rho^3 (1-p)^5 = \frac{1}{25} {}^8C_5 p^5 (1-p)^3$

$\Rightarrow \quad (1-p)^2 = \frac{1}{25} p^2$

$\Rightarrow \quad \left(\frac{p}{1-p}\right)^2 = (5)^2 \quad \Rightarrow \quad \frac{p}{1-p} = 5$

$\Rightarrow \quad p = \frac{5}{6} = \frac{m}{n}$

$\therefore \quad m + n - 5 = 5 + 6 - 5 = 6$

Q42 (9) - Integer Type

Let E_1 be the event that coin A is selected,

E_2 be the event that coin B is selected

E_3 be the event that coin C is selected, and F be the event of getting 2 heads and one tail on tossing coin three times.

$\therefore P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$

$P(F/E_1) = {}^3C_2 \left(\frac{1}{2}\right)^2 \cdot \frac{1}{2} = 3 \cdot \left(\frac{1}{2}\right)^3 = \frac{3}{8}$

$P(F/E_2) = {}^3C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right) = 3 \cdot \frac{4}{27} = \frac{4}{9}$

$P(F/E_3) = {}^3C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) = 3 \cdot \frac{2}{27} = \frac{2}{9}$

Now, it is given that $P(E_1/F) = \frac{k}{25}$

$\Rightarrow \quad \frac{P(E_1) \cdot P(F/E_1)}{[P(E_1) \cdot P(F/E_1) + P(E_2) \cdot P(F/E_2) + P(E_3) P(F/E_3)]} = \frac{k}{25}$

$\Rightarrow \frac{\frac{1}{3} \cdot \frac{3}{8}}{\frac{1}{3} \cdot \frac{3}{8} + \frac{1}{3} \cdot \frac{4}{9} + \frac{1}{3} \cdot \frac{2}{9}} = \frac{k}{25}$

$\Rightarrow \frac{\frac{3}{8}}{\frac{3}{8}+\frac{4}{9}+\frac{2}{9}}=\frac{k}{25}$

$\Rightarrow \frac{\frac{3}{8}}{\frac{27+48}{72}}=\frac{k}{25}$

$\Rightarrow \frac{3}{8}\times\frac{72}{75}=\frac{k}{25}$

$\Rightarrow \frac{9}{25}=\frac{k}{25}$

$\therefore k=9$

Q43 (1) - Integer Type

Let E_1, E_2, E_3 and A be the events defined as follows

E_1 : Bolt is manufactured by machine A.

E_2 : Bolt is manufactured by machine B. E_3 : Bolt is manufactured by machine C.

A : Bolt is defection.

Then, $\quad P(E_1)=\frac{25}{100}, P(E_2)=\frac{35}{100}, P(E_3)=\frac{40}{100}$

$P\left(\frac{A}{E_1}\right)=\frac{5}{100}, P\left(\frac{A}{E_2}\right)=\frac{4}{100}, P\left(\frac{A}{E_3}\right)=\frac{2}{100}$

$\therefore$ Required probability $=P(E_2/A)=\frac{P(E_2)P\left(\frac{A}{E_2}\right)}{\sum_{i=1}^{n}P(E_i)P\left(\frac{A}{E_i}\right)}$

$$\frac{\frac{35}{100}\times\frac{4}{100}}{\frac{25}{100}\times\frac{5}{100}+\frac{35}{100}\times\frac{4}{100}+\frac{40}{100}\times\frac{2}{100}}$$

$$=\frac{140}{125+140+80}=\frac{140}{345}=\frac{28}{69}$$

But given that, $\frac{m}{n}=\frac{28}{69}$

$\Rightarrow \quad \left|\frac{2m-n}{13}\right|=\left|\frac{2\times28-69}{13}\right|=1$

Q44 (2) - Integer Type

Let E_1, E_2 and A be the events defined as follow:

E_1 : Bag A chosen.

E_2 : Bag B chosen.

A: Ball is red.

Now,

$P(E_1)=P(E_2)=\frac{1}{2}$

$P\left(\frac{A}{E_1}\right)=\frac{3}{5}, P\left(\frac{A}{E_2}\right)=\frac{5}{9}$

By Baye's theorem, we have

$P(E_2/A)=\frac{P(E_2)P(A/E_2)}{P(E_1)P(A/E_1)+P(E_2)P(A/E_2)}$

$= \frac{\frac{1}{2}\times\frac{5}{9}}{\frac{1}{2}\times\frac{3}{5}+\frac{1}{2}\times\frac{5}{9}} = \frac{\frac{5}{9}}{\frac{3}{5}+\frac{5}{9}} = \frac{5}{9}\times\frac{45}{52} = \frac{25}{52}$

But it is given that, $\frac{m}{n} = \frac{25}{52}$

$\Rightarrow \quad |2m - n| = |50 - 52| = 2$

Q45 (5) - Integer Type

Let E_1, E_2, E_3 and A be the event defined as follow:

E_1 : Examinee guesses the answer.

E_2 : Examinee copies the answer.

E_3 : Examinee knows the answer.

A : Examinee answer correctly. We have, $P(E_1) = \frac{1}{3}P(E_2) = \frac{1}{6}$

Since, E_1, E_2 and E_3 are mutually exclusive and exhaustive events.

$P(E_1) + P(E_2) + P(E_3) = 1$

$\Rightarrow P(E_3) = 1 - [P(E_2) + P(E_1)]$

$= 1 - \left(\frac{1}{3} - \frac{1}{6}\right) = \frac{1}{2}$

If E_1 has already occurred, then the examinee guesses. Since, there are four choices out of which only one is correct, therefore the probability that he answers correctly given that he has made a guess is $\frac{1}{4}$ i.e. $P\left(\frac{A}{E_1}\right) = \frac{1}{4}$.

It is given that, $P\left(\frac{A}{E_2}\right) = \frac{1}{8}$ and $P\left(\frac{A}{E_3}\right) = 1$

By Baye's theorem, we have

$P\left(\frac{E_3}{A}\right) = \frac{P(E_3)P(A/E_3)}{P(E_1)P(A/E_1)+P(E_2)P(A/E_2)+P(E_3)P(A/E_3)}$

$= \frac{\frac{1}{2}\times 1}{\frac{1}{3}\times\frac{1}{4}+\frac{1}{6}\times\frac{1}{8}+\frac{1}{2}\times 1} = \frac{24}{29}$

But $\quad \frac{m}{n} = \frac{24}{29}$

$\therefore \quad |m - n| = |24 - 29| = 5$

Q46 (3) - Integer Type

Here, $n(S) = 216$

$x_1 =$ Number appearing on first die

$x_2 =$ Number appearing on second die

$x_3 =$ Number appearing on third die

$x_1 + x_2 + x_3 \leq 10 \quad [\because x_1, x_2, x_3 \in [1, 6]]$

$\Rightarrow \quad x_1 + x_2 + x_3 \leq 7 \quad$ [after giving 1 each to x_1, x_2, x_3]

$\therefore \quad x_1 + x_2 + x_3 + X = 7 \quad$ [adding X as a false beggar]

Total number of solution $= {}^{7+3}\mathrm{C}_3$

$= {}^{10}\mathrm{C}_3 = 120$

Now, number of solutions when any one of x_1, x_2, x_3 takes the value 7 is

$x_1 + x_2 + x_3 + X = 1$
${}^{1+3}C_3 = 4$

$\therefore$ Total number of ways $= {}^{10}C_3 - {}^{4}C_3 \times {}^{3}C_1$

$= 108$

$\therefore$ Required probability $= \frac{108}{216} = \frac{1}{2}$

Now, $\quad m + n = 3$